

Orbital Rotation Gradient and Hessian for Exponential Parameterization (RDMFT-style, expressed via the Generalized Fock Matrix)

Conventions

One-particle and two-particle reduced density matrices:

$$D_{ab} = \langle 0 | a_a^\dagger a_b | 0 \rangle, \quad (1)$$

$$\Gamma_{pqrs} = \frac{1}{2} \langle 0 | a_p^\dagger a_q^\dagger a_s a_r | 0 \rangle. \quad (2)$$

Two-electron integrals in physicist's notation, $\langle ik|jl \rangle$.

Hamiltonian:

$$\hat{H} = \hat{H}_1 + \hat{H}_2, \quad \hat{H}_1 = \sum_{pq} h_{pq} a_p^\dagger a_q, \quad \hat{H}_2 = \frac{1}{2} \sum_{uvwx} \langle uv|wx \rangle a_u^\dagger a_v^\dagger a_x a_w. \quad (3)$$

Orbital rotation generator (anti-Hermitian): $\hat{\kappa} = \sum_{pq} \kappa_{pq} a_p^\dagger a_q$, with $\kappa_{qp} = -\kappa_{pq}^*$.

Generalized Fock matrix

$$F_{qp} = \sum_r D_{rq} h_{rp} + 2 \sum_{rst} \Gamma_{srtq} \langle sr|tp \rangle. \quad (4)$$

Gradient

$$g_{pq} = F_{qp} - F_{pq}^*. \quad (5)$$

Hessian

Energy expansion:

$$E(\boldsymbol{\kappa}) \approx E(\mathbf{0}) + \sum_{pq} \kappa_{pq} g_{pq} + \frac{1}{2} \sum_{pq,rs} \kappa_{pq} \kappa_{rs} G_{pq,rs} + \dots, \quad G_{pq,rs} = \langle 0 | [[\hat{H}, a_p^\dagger a_q], a_r^\dagger a_s] | 0 \rangle. \quad (6)$$

One-electron contribution

$$G_{pq,rs}^{(1)} = \delta_{qr} \sum_u h_{up} D_{us} - h_{sp} D_{rq} - h_{qr} D_{ps} + \delta_{sp} \sum_v h_{qv} D_{rv}. \quad (7)$$

Two-electron contribution

$$\begin{aligned}
G_{pq,rs}^{(2)} = & + 2 \delta_{rq} \sum_{uvt} \langle uv|pt \rangle \Gamma_{uvst} - 2 \sum_{uv} \langle uv|pr \rangle \Gamma_{uvsq} \\
& + 2 \sum_{ut} \langle us|pt \rangle \Gamma_{ruqt} - 2 \sum_{ut} \langle su|pt \rangle \Gamma_{ruqt} \\
& + 2 \sum_{ut} \langle uq|rt \rangle \Gamma_{pust} - 2 \sum_{uw} \langle uq|wr \rangle \Gamma_{pusw} \\
& + 2 \sum_{wt} \langle sq|wt \rangle \Gamma_{rpwt} - 2 \delta_{sp} \sum_{uwt} \langle uq|wt \rangle \Gamma_{ruwt}.
\end{aligned} \tag{8}$$

Full Hessian, Fock-matrix form

The δ_{qr} and δ_{sp} terms of $G^{(1)}$ and $G^{(2)}$ combine exactly into F_{sp} and F_{rq}^* , leaving six two-electron “exchange-type” terms that do not reduce to F :

$$\begin{aligned}
G_{pq,rs} = & \delta_{qr} F_{sp} + \delta_{sp} F_{rq}^* - h_{sp} D_{rq} - h_{qr} D_{ps} \\
& - 2 \sum_{uv} \langle uv|pr \rangle \Gamma_{uvsq} + 2 \sum_{ut} \langle us|pt \rangle \Gamma_{ruqt} - 2 \sum_{ut} \langle su|pt \rangle \Gamma_{ruqt} \\
& + 2 \sum_{ut} \langle uq|rt \rangle \Gamma_{pust} - 2 \sum_{uw} \langle uq|wr \rangle \Gamma_{pusw} + 2 \sum_{wt} \langle sq|wt \rangle \Gamma_{rpwt}
\end{aligned} \tag{9}$$

Real/imaginary directional blocks

For a complex (relativistic, spin-orbit-coupled) spinor basis, κ_{pq} for an independent pair $p > q$ is a free complex number, $\kappa_{pq} = t_{pq} + iy_{pq}$ ($\kappa_{qp} = -t_{pq} + iy_{pq}$, enforcing $\kappa_{qp} = -\kappa_{pq}^*$). The full real Hessian over these $2 \times (\text{number of pairs})$ real parameters therefore has three independent blocks between any two distinct pairs (p, q) and (r, s) , all obtainable from the SAME four raw terms $G_{pq,rs}$, $G_{pq,sr}$, $G_{qp,rs}$, $G_{qp,sr}$ by substituting $\kappa_{pq} = t + iy$, $\kappa_{qp} = -t + iy$, $\kappa_{rs} = t' + iy'$, $\kappa_{sr} = -t' + iy'$ into $\frac{1}{2} \sum \kappa_{pq} \kappa_{rs} G_{pq,rs}$ and reading off the coefficient of each real product:

$$\text{Hess}_{pq,rs} = \frac{\partial^2 E}{\partial t_{pq} \partial t_{rs}} = G_{pq,rs} - G_{pq,sr} - G_{qp,rs} + G_{qp,sr}, \tag{10}$$

$$\text{Hess}_{pq,rs}^{yy} = \frac{\partial^2 E}{\partial y_{pq} \partial y_{rs}} = -(G_{pq,rs} + G_{pq,sr} + G_{qp,rs} + G_{qp,sr}), \tag{11}$$

$$\text{Hess}_{pq,rs}^{ty} = \frac{\partial^2 E}{\partial t_{pq} \partial y_{rs}} = i(G_{rs,pq} + G_{rs,qp} - G_{sr,pq} - G_{sr,qp}). \tag{12}$$

The first two are the $(+, -, -, +)$ and $(-, -, -, -)$ sign patterns over the four raw terms with (p, q) in the flipping (first) slot; the third is the same idea but with the *other* pair, (r, s) , in the flipping slot – not (p, q) , which is easy to get backwards by analogy with the first two (an earlier draft of this derivation did exactly that, and was only caught by the finite-difference check below). The fourth possible direction, $\text{Hess}_{pq,rs}^{yt} = \partial^2 E / \partial y_{pq} \partial t_{rs}$, is not a new formula: by Schwarz’s theorem, $\text{Hess}_{pq,rs}^{yt} = \text{Hess}_{rs,pq}^{ty}$.

All three are real for physical (Hermitian h , $\langle \cdot | \cdot \rangle$, D , Γ) input by the same argument: $K_{pq} = a_p^\dagger a_q - a_q^\dagger a_p$ and $K'_{rs} = i(a_r^\dagger a_s + a_s^\dagger a_r)$ are both anti-Hermitian, so $[[\hat{H}, K_{pq}], K'_{rs}]$ is Hermitian and its expectation value is real, regardless of which of the three sign patterns above it corresponds to.

Status note (updated). $G_{pq,rs}^{(1)}$, the gradient g_{pq} , $\text{Hess}_{pq,rs}$, and $\text{Hess}_{pq,rs}^{yy}$ have all been independently verified: by more than one derivation route, by an explicit small-Fock-space numerical check against a general (non-idempotent, non-Wick-derivable) random 1-/2-RDM, and by finite-difference checks against converged NON_REL/C4.DHF SCF data. $\text{Hess}_{pq,rs}^{ty}$ (and hence $\text{Hess}_{pq,rs}^{yt}$ via Schwarz) has since also been derived and confirmed the same way – a genuine mixed-direction (real step on one pair, imaginary step on the other) 2D finite difference against water/STO-3G’s converged AND a deliberately non-converged (1 SCF iteration) C4.DHF density, matching to $O(\text{step}^2)$ in both cases (implementation: `Hessian_opt/HartreeExchangeHessian.h`’s `hartreeExchangeHessianElementMixed`, the Hartree/exchange-ansatz ”cheap” path only – the fully general (`GeneralizedHessian.h`) counterpart has not been added). The only remaining unimplemented piece for a full $2N(N-1)$ -real-parameter orbital-rotation Hessian is the general (non-Hartree/exchange-ansatz) 2-RDM version of the mixed block, and assembling all three blocks into one joint matrix for an actual Newton-Raphson/NEO step (currently only $\text{Hess}_{pq,rs}$ alone is ever assembled into a diagonalizable matrix, for the minimum/saddle-point diagnostic in `main.cpp`’s `HESSIAN_4C`).

1 Cheap ansatz for PNOF functionals: $G_{pq,rs}$ in terms of `two_rdm_H`, `two_rdm_X`, and the pair terms L1/L2

The ansatz

Every PNOF functional this project implements (`Hessian_opt/PnofFock.h`) is handled by the SAME cheap diagonal- D 2-RDM ansatz already used for the ordinary Hartree/exchange gradient (`Hessian_opt/HartreeExchangeGradient.h`), extended with two further ”pair” terms:

$$\Gamma_{pqrs} = \frac{1}{2} \left[H_{pq} \delta_{pr} \delta_{qs} - X_{pq} \delta_{ps} \delta_{qr} \right] + \delta_{q,\bar{p}} \delta_{s,\bar{r}} \left[L_{pr}^1 + L_{\bar{p}\bar{r}}^2 \right], \quad (13)$$

where $H_{pq} \equiv \text{two_rdm_h}(p, q)$, $X_{pq} \equiv \text{two_rdm_x}(p, q)$, $L_{pq}^1 \equiv \text{two_rdm_l1}(p, q)$, $L_{pq}^2 \equiv \text{two_rdm_l2}(p, q)$ (all real, independent of the orbital basis being real or complex), and $\bar{p} \equiv \text{pair_of}(p)$ is p ’s Kramers/spin partner. Equation (13) says: the plain Hartree/exchange part fires at $(p, q, r, s) = (p, q, p, q)$ and (p, q, q, p) as usual; the pair part fires ONLY when $q = \bar{p}$ AND $s = \bar{r}$ (i.e. the first PAIR (p, q) and the third/fourth PAIR (r, s) are each a Kramers/spin pair themselves), where it evaluates to $L_{pr}^1 + L_{\bar{p}\bar{r}}^2$ – BOTH L^1 and L^2 contribute at that SAME tensor entry, evaluated at different index pairs (L^1 at (p, r) , L^2 at $(p, \bar{r})$), exactly as

`HartreeExchangeGradient.cpp`’s own F_{pq} derivation already uses them. For a PNOF functional, $H_{pq} = X_{pq}$ (uniform across every Kramers/spin bar-combination of a geminal pair, `doc/rel_pnofs.tex`’s own `eq:coulomb-elems`) and $L_{pq}^1 = -L_{pq}^2 = \pm \Pi_{\text{rep}(p), \text{rep}(q)}/2$ (sign $+$ at matching bar-parity, $-$ at mismatched) – see `PnofFock.h`’s own header comment for the full derivation from `doc/rel_pnofs.tex`’s Eqs. 98c–f.

Hartree/exchange contribution to $G_{pq,rs}$

Substituting only the H/X part of Eq. (13) into the boxed $G_{pq,rs}$ of Eq. (9) (already implemented, `Hessian_opt/HartreeExchangeHessian.h`’s `hartreeExchangeHessianElement`, before this section’s own L1/L2 extension) collapses every double sum there to a single one, exactly as

`hartreeExchangeFockMatrix`'s own gradient derivation already does for F_{pq} :

$$\begin{aligned}
G_{pq,rs}^{\text{HX}} = & \delta_{qr} F_{sp} + \delta_{sp} F_{rq}^* - \delta_{qr} h_{sp} D_{rq} - \delta_{sp} h_{qr} D_{ps} \\
& + \delta_{qr} \sum_u H_{ru} [\langle us|pu \rangle - \langle su|pu \rangle] + \delta_{sp} \sum_u H_{pu} [\langle uq|ru \rangle - \langle uq|ur \rangle] \\
& + \langle sq|pr \rangle [X_{rq} - H_{sq} + X_{ps} - X_{rp}] + \langle sq|rp \rangle [H_{rp} + X_{qs} - X_{ps} - X_{rq}].
\end{aligned} \tag{14}$$

Pair (L1/L2) contribution to $G_{pq,rs}$

Substituting only the pair part of Eq. (13) into the SAME six two-electron sums of the boxed $G_{pq,rs}$ and collapsing each double sum via its own $\delta_{q,\bar{p}}\delta_{s,\bar{r}}$ -type constraint (four of the six collapse completely to a SINGLE eri times a SINGLE $L^1 + L^2$ coefficient; the other two collapse only conditionally, leaving one remaining sum, exactly when the OUTER indices already satisfy the pair condition) gives a genuinely new contribution, not obtainable from Eq. (14) alone:

$$\begin{aligned}
G_{pq,rs}^{\text{L}} = & -2 [q = \bar{s}] \sum_u \langle u\bar{u}|pr \rangle [L_{us}^1 + L_{uq}^2] \\
& + 2 \langle \bar{r}s|p\bar{q} \rangle [L_{rq}^1 + L_{r\bar{q}}^2] - 2 \langle s\bar{r}|p\bar{q} \rangle [L_{rq}^1 + L_{r\bar{q}}^2] \\
& + 2 \langle \bar{p}q|r\bar{s} \rangle [L_{ps}^1 + L_{p\bar{s}}^2] - 2 \langle \bar{p}q|\bar{s}r \rangle [L_{ps}^1 + L_{p\bar{s}}^2] \\
& + 2 [p = \bar{r}] \sum_w \langle sq|w\bar{w} \rangle [L_{rw}^1 + L_{r\bar{w}}^2],
\end{aligned} \tag{15}$$

where $[\cdot]$ is the Iverson bracket (1 if the condition holds, 0 otherwise) – the first and last lines vanish identically unless the OUTER pair (q, s) or (p, r) already happens to satisfy the SAME Kramers/spin-pair relation the ansatz itself requires, in which case a single sum over the remaining free index survives; the middle two lines (four raw terms, pairwise sharing the SAME $L^1 + L^2$ coefficient) are UNCONDITIONAL, contributing to every (p, q, r, s) .

Combined result

$$G_{pq,rs} = G_{pq,rs}^{\text{HX}} + G_{pq,rs}^{\text{L}}, \tag{16}$$

and the real-step Hessian follows exactly as in Eq. (10) (this file's own $(+, -, -, +)$ antisymmetrization over the two pairs, applied to the COMBINED $G_{pq,rs}$ of Eq. (16), not to G^{HX} and G^{L} separately – though since the antisymmetrization is linear, applying it to each piece and adding is equivalent, which is exactly how `Hessian_opt/HartreeExchangeHessian.cpp`'s own implementation is structured, two independent raw-term helpers summed before/after antisymmetrizing).

Status

Implemented in `Hessian_opt/HartreeExchangeHessian.h`'s `hartreeExchangeHessianElement` (optional `pair_of/ two_rdm_l1/two_rdm_l2` arguments, default empty reducing to Eq. (14) alone) and wired into PNOF via `Hessian_opt/PnofHessian.h`. Only the REAL-step block $\text{Hess}_{pq,rs}$ (Eq. (10)) has this L1/L2 extension so far – $\text{Hess}_{pq,rs}^{yy}$ and $\text{Hess}_{pq,rs}^{ty}$ do not yet. Validated two ways: (i) against `GeneralizedHessian.h`'s fully general `generalizedOrbitalHessianElement`, fed an explicit dense Γ built from the SAME Eq. (13) ansatz, on random data (T = real and complex,

H/X/ L^1/L^2 deliberately unequal and unrelated to any occupation-number formula) – machine precision across every tested (p, q, r, s) quadruple; (ii) against a genuine $\mathbf{e}^\kappa$ finite-difference probe (`Hessian_opt/OrbitalRotationFiniteDifference.h`) on real, converged LiH/6-31G SCF data for BOTH the X2C and 4-component (C4_DHF) paths – machine precision ($\sim 10^{-14}$) there too, confirmed robust across multiple (p, q, r, s) quadruple choices including ones where (r, s) are (p, q) ’s own Kramers-bar-partners.

Before this L1/L2 extension existed, PNOF’s own `PnofFock.h` instead folded Π (the geminal pairing coefficient) directly into X_{pq} and used no L1/L2 terms at all. That simplification is EXACTLY energy- and Fock-equivalent for a Kramers-symmetric spinor basis (an exact integral identity, $\langle p\bar{p}|q\bar{q}\rangle = \langle pq|qp\rangle$ and $\langle p\bar{p}|\bar{q}q\rangle = -\langle \bar{p}q|q\bar{p}\rangle$, confirmed numerically to machine precision) but is NOT Hessian-equivalent: $G_{pq,rs}$ fixes all four indices explicitly, so it sees the difference between weight sitting at the X_{pq} tensor position versus the L_{pq}^1/L_{pq}^2 one, even though F_{pq} and E do not. This is why the L1/L2 extension above is necessary, not merely convenient, for a correct PNOF Hessian.